DATA ITEM DESCRIPTION

Title: Weapon Safety Fire Hazard Analysis

Number: DI-FACR-82415
AMSC Number: F10430
DTIC Applicable: No
Preparing Activity: 37 (AFSEC/SEWN)
Applicable Forms: N/A

Approval Date:20230929
Limitation: N/A
GIDEP Applicable: No
Project Number: FACR-2023-001

Use/relationship: A Weapon Safety Fire Hazard Analysis (FHA), aforementioned as FHA, is a technical evaluation performed by a nuclear weapons System Program Office (SPO) or contracted to a qualified professional by the SPO to document compliance with all the nuclear safety design certification requirements identified in the respective Certification Requirements Plan (CRP). The purpose of the FHA is to assess the risk, comprehensively and quantitatively, from a fire originating from within the fire areas of an Air Force facility. The assessment is concerned with the risk to weapons only and must ascertain if Air Force nuclear safety design criteria is met. The FHA shall provide an executive summary of the analytical results and conclusions. Additionally, the FHA describes the key systems of the fire suppression, the specific components, methods used, and concerns resulting from analyses performed as well as a statement regarding compliance or non-compliance with the nuclear safety design requirements in Air Force Manual (AFMAN) 91-118, Safety Design and Evaluation Criteria for Nuclear Weapon Systems, AFMAN 91-119, Safety Design Criteria for Nuclear Weapon Systems Software, and Department of the Air Force Instruction (DAFI) 91-101, Air Force Nuclear Weapons Surety Program.

This Data Item Description (DID) contains the format and content preparation information for the deliverable resulting from the work task described in the solicitation and contract. This DID is related to ***NPFC - DI-NUOR-81409,*** *Certification Requirements Plan (CRP)*, AFMAN 91-118, DAFMAN 91-119, and DAFI 91-101. (Copies of DAFIs, DAFMANs, and AFMANs are available online at http://www.e-publishing.af.mil/.)

Requirements:

1. Reference Documents: The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions, shall be as cited in the current issue of the DODISS at the time of the solicitation; or, for non DODISS-listed documents, as stated herein.

2. Format: The Weapon Safety FHA shall be presented in the format specified in section 4 (General Outline) of this document.

3. Content: The FHA consists of two sections, the Design FHA and the Post-Construction FHA.

    3.1. The Design FHA must be initiated early in the design phase to ensure safety design criteria is being properly incorporated into the design of the facility. The Design FHA shall include an assessment of the risk from fire and other related hazards (e.g. direct flame impingement, hot gases, smoke migration) to the weapons. The assessment shall include the fire model analysis and evaluation. Fire models will be based on fire scenarios determined by AFSEC/SEWN and

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the Special Program Office (SPO), which will be based on facility features and Concept of Operations (CONOPS). AFSEC/SEWN must agree with model scenarios. The design FHA shall comprehensively, qualitatively, and quantitatively assess the following:

3.1.1. The probability of fires within each new or modified fire area within a facility with the potential to impact the weapons.

3.1.2. The effects of possible fires and related perils in relation to the fire safety features' capabilities to ensure fire events can be safely controlled and stabilized for each new and modified fire area that could potentially impact the weapons.

3.1.3. When performing an evaluation assume all potentially vulnerable systems will be damaged within fire areas except water-filled steel pipes, tanks, and similar components of superior structural integrity with welded fittings and adequate pressure relief. Passive fire protection features, such as blank fire-rated walls, floors, ceilings, and continuous fire-rated cable wraps, are assumed to remain viable. If redundant automatic fire protection systems are provided in an area, only the one system or component that causes the most vulnerable condition need be assumed unavailable.

3.1.4. The thermal parameters, through modeling of fire events for each new and modified fire area that may contain a weapon, then compare the results to the weapon screening values.

3.1.4.1. Fire Modeling will:

3.1.4.1.1. Model material established by agreement between the relevant stakeholders, including AFSEC/SEWN.

3.1.4.1.2. Model dimensions established by agreement between relevant stakeholders, including AFSEC/SEWN.

3.1.4.1.3. Run models until temperatures of modeled material cease to increase.

3.1.5. The impact of the facility fire protection system on nuclear criticality following a fire event and blast event (if applicable) for each new or modified area.

3.2. The Post-Construction FHA will must comprehensively and quantitatively assess the fire originating from within individual fire areas to ascertain if the Air Force weapon safety objectives are being met by ensuring the validity of the Design FHA. The construction of the facility must not deviate from the design to the extent that impacts fire modeling and assumptions. Post-Construction equipment and operational programs must be evaluated for potential fire hazards that were not considered in the Design FHA. Post-Construction FHA must include the following:

3.2.1. Verify as-built construction is consistent with design specifications and drawings. Any discrepancy between construction and design documents that could potentially, directly, or indirectly, impact fire modeling assumptions, parameters, or results must be addressed through analysis and possibly remodeling, depending on the severity of impact to the model.

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3.2.2. Provide an assessment of severity impact to fire modeling for any identified discrepancy. The final determination of what is required to redress any discrepancies will be made by AFSEC/SEWN and AFGSC.

3.2.3. Recommendations for the correction or mitigation of non-compliance. The recommendations must identify specific mitigation strategies for the non-compliance. For any non-compliance, the Post-Construction FHA will include a risk assessment of the non-compliance.

3.2.4. Evaluate Post-Construction equipment and operational programs that were not evaluated in the Design FHA to ensure no new fire hazards were introduced to facility areas where weapons may be.

3.2.5. Recommendations for improvement of features in compliance. For recommendations of improvement for features in compliance, a risk assessment is not required.

4. General Outline: The Design FHA will provide:

1. Executive Summary

1.1 Purpose and brief description of FHA

1.2 Brief description of fire protection system

1.3 Brief description of findings and recommendations

2. Introduction

2.1 Purpose

2.2 Requirements

2.3 Approach and assumptions

2.4 Limitations

2.5 Facility use and function

2.5.1 General

2.5.2 Operations and processes

2.5.3 Lightning protection system

2.5.4 Nuclear weapons side-flash protection

2.5.5 Electrical power systems

2.5.6 Fire protection systems

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2.5.7 Hoist, cranes, and similar devices

2.5.8 Blast containment/isolation features

2.5.9 Electromagnetic radiation environment

3. Fire protection

3.1 Water supply and distribution system

3.2 Fire suppression

3.3 Fire alarm system

3.4 Relevant Non-Analytical Information

4. Hazards

4.1 Fixed fire hazards

4.2 Transient fire hazards

4.3 Natural phenomena hazards

5. Fire scenario selection and analysis

5.1 Scenario selection rationale

5.2 Description of scenario

5.3 Analysis of fire scenario

6. Fire risk analysis

6.1 Identification and protection of critical process equipment

6.2 Maximum credible fire loss

6.3 Maximum possible fire loss

7. Findings and recommendation

7.1 Closed findings

7.2 Open findings

7.3 AFMAN 91-118/119 deficiencies

7.4 Recommendations

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7.5 Risk estimation

5. General Outline: The Post-Construction FHA will provide:

1. Executive summary

1.1 Purpose and brief description of FHA

1.2 Brief description of fire protection system

1.3 Brief description of findings and recommendations

2. Introduction

2.1 Purpose

2.2 Requirements

2.3 Approach and assumptions

2.4 Limitations

2.5 Facility use and function

2.5.1 General

2.5.2 Operations and processes

2.5.3 Lightning protection system

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2.5.5 Electrical power systems

2.5.6 Fire protection systems

2.5.7 Hoist, cranes, and similar devices

2.5.8 Blast containment/isolation features

2.5.9 Electromagnetic radiation environment

3. Evaluation

3.1 Validity of Design FHA

3.1.1 Consistency of construction with design specifications, and drawings

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3.2 Post-construction equipment

3.3 Operational programs

4. Findings and recommendations

4.1 Closed findings

4.2 Open findings

4.3 AFMAN 91-118/119 deficiencies

4.4 Recommendations

4.5 Risk estimation

End of DI-FACR-82415

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